GLENDA HUI EN TAN

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EDUCATION

Carnegie Mellon University

Doctor of Philosophy in Electrical and Computer Engineering Aug 2026–present

Bachelor of Science in Artificial Intelligence (3.91 GPA) Aug 2023–May 2026

Relevant Coursework: Generative AI, Autonomous Agents, ML for Radar, Mobile and Pervasive Computing, Computer Vision, Speech Technology for Conversational AI, Design of AI Products

EXPERIENCE

Research Assistant, *Reliable Autonomous Systems Lab & AI-CARING Institute CMU* May 2024–present

Advisors: Prof. Reid Simmons, Prof. Zackory Erickson and Prof. Pragathi Praveena (CMU Robotics Institute)

• Developed an AI-driven tracking and localization system to assist older adults in locating kitchen items.
• Engineered an object detector leveraging Grounded SAM 2 and prompt engineering to identify 20+ kitchen items across diverse meal-preparation scenarios.
• Designed a scalable registration pipeline using RGB-D cameras to enable system deployment across varied kitchen environments.
• Evaluated system performance on real-world video datasets of 8 older adult participants preparing meals.
• Recognition & Funding: Awarded the CMU Summer Undergraduate Research Fellowship (SURF 2025).

Deep Learning Research Intern, *Defence Science Organization (DSO) National Laboratories Singapore* July 2021–May 2022

Advisor: Dr. Bingquan Shen (DSO Information Division)

• Conceptualized and implemented a Transformer-based predictive model to forecast future high-infectivity COVID-19 viral variants.
• Achieved 90% accuracy in protein-protein interaction prediction by incorporating Masked Language Modeling, Sharpness-Aware Minimization, and custom data augmentation techniques.
• Published as First Author: "Predicting More Infectious Virus Variants for Pandemic Prevention through Deep Learning" (CMLA 2022 & IJAIA 2022).
• Awards: Singapore Science and Engineering Fair (SSEF) 2022 Gold Award; Regeneron ISEF 2022 Atlanta USA Finalist.

Machine Learning Research Intern, *DSO National Laboratories Singapore* May 2020–June 2021

Advisor: Dr. Bingquan Shen (DSO Information Division)

• Architected a deep convolutional neural network for predicting colorectal cancer risks from stool images with 94% accuracy.
• Addressed dataset scarcity by fine-tuning a DiffAugment StyleGAN2 pipeline to synthesize realistic, high-fidelity stool images.
• Published as First Author: "Stool Recognition for Colorectal Cancer Detection through Deep Learning" (IEEE ICMI 2025).
• Awards: SSEF 2021 Gold Award; Global Youth & Science Technology Bowl Hong Kong (GYSTB) Grand Prize (Bronze); Singapore University of Technology and Design Research & Innovation Award in Healthcare.

TEACHING EXPERIENCE

Head Teaching Assistant, *Concepts of AI Course [07-180] CMU School of Computer Science* Spring 2025-May 2026

• Led a team of 7 Teaching Assistants overseeing course logistics, grading, and recitations for 100+ freshman CS students.
• Directed curriculum delivery covering core AI domains, including generative AI, reinforcement learning (RL), and AI ethics.
• Authored original assignments, exams, and a custom 2048 RL agent competition to promote hands-on algorithm implementation.
• Conducted weekly recitations and office hours, achieving a 5.0/5.0 TA evaluation rating from students.

PUBLICATIONS

1. **G. Tan**, K. Goh and B. Shen, “Stool Recognition for Colorectal Cancer Detection through Deep Learning,” *IEEE 4th International Conference on Computing and Machine Intelligence (ICMI) Michigan USA*, April 2025, doi: 10.1109/ICMI65310.2025.11141177 | *IEEE Xplore* | *arXiv*, October 2024.
2. C. Chan and **G. Tan**, “AI vs Human Creativity: Investigating the Effectiveness of AI-Generated Advertisements”, *National Conference of Undergraduate Research (NCUR) Pittsburgh USA*, pp. 265-266, April 2025.
3. **G. Tan** and C. Szalkowski-Ference, “The Role of Artificial Intelligence in Art Restoration,” *WOVEN: An Interdisciplinary Journal of Dietrich College CMU*, Issue 4, pp. 11-26, Spring 2024.

4. J. Kubica, R. Kumar, **G. Tan** et al, "The fourth annual Carnegie Mellon Libraries hackathon for biomedical data management, knowledge graphs, and deep learning," *BioHackrXiv*, Nov 2023, doi: 10.37044/osf.io/k4rh6

5. **G. Tan**, T. E. Koay and B. Shen, "Predicting More Infectious Virus Variants for Pandemic Prevention through Deep Learning," *International Journal of Artificial Intelligence and Applications (IJAIA)*, vol. 13, no. 4, pp. 33-54, July 2022, doi: 10.5121/ijaia.2022.13403 | *4th International Conference on Machine Learning and Applications (CMLA) Copenhagen Denmark*, vol. 12, no. 11, pp. 17-31, June 2022, doi: 10.5121/csit.2022.121102 | *arXiv*, May 2022.

HONORS AND AWARDS

CMU Senior Leadership Recognition Award	Spring 2026
Phi Beta Kappa CMU Chapter Honoree	Spring 2026
CMU Andrew Carnegie Society (ACS) Scholar	2025-2026
CMU Office of Undergraduate Research and Scholar Development Presentation Award	Spring 2025
CMU School of Computer Science: Dean's List with High Honors	Fall 2025, Fall 2024, Spring 2024, Fall 2023
Agency for Science, Technology and Research Science Award, Singapore	2021–2022
RoboCup Asia-Pacific: 1st Place in Rescue Line category, Aichi Japan	2021

EXTRACURRICULARS AND LEADERSHIP ACTIVITIES

Conference Reviewer

- *IEEE Spoken Language Technologies Conference, Sicily, Italy* — July 2026
- *14th European Alliance for Innovation International Conference: ArtsIT, Interactivity & Game Creation, Dubai, UAE* — Aug 2025

CMU Robotics Club — Pittsburgh, Pennsylvania

Lead Programmer for BB-8 Droid Project (a collaboration with CMU Star Wars Club) — Fall 2024–present

- Directing a 20-member team in building control systems and programming Arduino circuits for a life-sized, functional BB-8 droid.
- 3D-printed custom mechanical components and assembled the main spherical chassis.
- Architecting real-time embedded control software to manage locomotion and balance.

CMU Star Wars Club — Pittsburgh, Pennsylvania

Co-founder, President (2023-2025) and Lightsaber Grandmaster (2025-2026) — Fall 2023–present

- Co-founded the CMU Star Wars Club with two friends in Fall 2023 to build a community of Star Wars fans across the CMU galaxy.
- Recruited 354 members from Fall 2023 to Spring 2026; led schoolwide events like May the 4th Celebration & Rise Against Hunger.
- Lightsaber training instructor and performance choreographer for CMU SCS Day and ScottyCon.

CMU AI in Action Seminar Series — Pittsburgh, Pennsylvania

Organizing Team Member — Fall 2024–May 2026

- AI in Action invites AI leaders to share their insights with the CMU community, speakers include Nobel Laureate Geoffrey Hinton.
- Editor of an original video series showcasing CMU AI professors' research and views on the future of AI in society (Prof. Reid Simmons, Prof. Ding Zhao, Prof. Zico Kolter and Prof. Graham Neubig), published videos on the CMU SCS YouTube channel.

PROJECTS

- *HaikuS2S:* Built a cascaded speech system integrating an ASR model, LLM, and TTS model fine-tuned on haiku prosody.
- *Hair Diffuser*: Fine-tuned Stable Diffusion Inpainting generative model with DreamBooth LoRA to simulate users in new hairstyles.
- *Autonomous Greenhouse*: Programmed autonomous greenhouse with light monitor & ResNet-50 foliage classifier to grow radishes.
- *Aurebesh Decoder*: Fine-tuned vision transformer & ResNet-50 to translate messages in the Star Wars language Aurebesh to English.

SKILLS

ML Models: LLMs, Diffusion Models, Transformers, GANs, ASR, TTS, VLMs, CNNs
DL/Agent Frameworks: PyTorch, HuggingFace, OpenCV, NumPy, Pandas, Wandb, LangChain, LangGraph, CrewAI, AG2
Programming Languages: Python, C, JavaScript, R, C++, C#, SML, Java, Swift, HTML/CSS
Other Software: Arduino, React Native, Node.js, Unity, Unreal Engine 4, LaTeX, XCode, Android Studio
Research: Academic Writing, Literature Review, Conference Review, Teaching/Mentoring, Community Outreach
Languages: English (native), Chinese (proficient), Spanish (intermediate), Japanese (elementary)